

Introduction to FFLO Superconductivity

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Outline

BCS

FFLO

Experiments

Pairing symmetry

$$V_{\text{eff}}(r) < 0$$

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$$\rightarrow \psi_s(1, 2) = \psi_s(2, 1)$$

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$$\langle V \rangle_t > \langle V \rangle_s$$

Cooper Pair

$$\mathcal{H} = \sum_{k,\sigma} \epsilon_k \hat{n}_{k,\sigma} + \sum_{k,k'} V_{k,k'} \hat{c}_{k\uparrow}^\dagger \hat{c}_{-k\downarrow}^\dagger \hat{c}_{k'\uparrow} \hat{c}_{-k'\downarrow}$$

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$$V^{-1} = \sum_{|k'| > k_F} (E - 2\epsilon_k)^{-1}$$

Band Gap

$$|BCS\rangle = \prod_{k > k_{min}} (u_k + v_k \hat{c}_{k\uparrow}^\dagger \hat{c}_{-k\downarrow}^\dagger) |O\rangle$$

Zeeman Splitting

hello

Fermi shell

hello

New Auxiliary Field

hello

Hard To Measure

hello

EXP 1

hello

EXP 2

hello